

PRANAV VUDUMULA

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EDUCATION

University of Maryland – College Park

B.S. Computer Science (Machine Learning Track); Mathematics Major; Robotics Minor

College Park, MD

Aug. 2023 – May 2027

Relevant Coursework: Robotics Programming, Computer Vision, Machine Learning, Human-Robot Interaction, Data Structures & Algorithms, Applied Statistics, Cloud Computing

EXPERIENCE

Undergraduate Research Assistant, University of Maryland — Project Agentic Tangibles

Faculty-supervised robotics research, with PhD researcher

College Park, MD

Summer 2026 – Present

- Calibrated and debugged a BNO055 IMU-based orientation/face-detection system under real deployment conditions, engineering hysteresis dead zones and debounce logic to filter sensor noise and eliminate false state transitions from an off-axis sensor mount.
- Engineered embedded C++ firmware (ESP32-S2) for a multi-node sensor network: persistent TCP streaming at 5Hz, OTA updates via ArduinoOTA, and NVS-backed device identity for fleet-wide firmware pushes without overwriting per-board identity.
- Designed a thread-safe Python data-collection server (rolling per-node buffers) and an LLM command-dispatch pipeline constrained to a whitelisted JSON schema, preventing hallucinated commands from reaching physical hardware; targeting a CHI 2027 co-authorship.

Software Engineer Intern, Socratics.ai

San Francisco, CA

May 2025 – Aug 2025

- Architected an LLM-based assessment platform (Gemini, RAG over a 1,000+ question bank) and scalable ETL/vector-search pipelines processing 5M+ records at sub-200ms latency, reducing manual grading effort by 60%.
- Built CI/CD workflows with GitHub Actions and Docker on Google Cloud Vertex AI, scaling to 10,000+ monthly sessions with a sub-5% rollback rate while increasing release velocity by 25%.
- Owned features end-to-end through code review, on-call rotations, and agile sprints from design through production deployment.

Data Science Intern, Tetra Science

Remote

Jun 2024 – Aug 2024

- Refactored Python automation scripts for scientific data pipelines and resolved failing end-to-end tests, increasing reliability by 30%, raising test coverage by 20%, and reducing CI/CD failures by 40%.
- Developed a manifest-update tool automating artifact compatibility for 80+ assets, saving the engineering team 15+ hours per week, while ensuring 100% data-integrity compliance with platform standards.

PROJECTS

Embodied Reasoning Agent — Vision-Language Robot Navigation

Summer 2026

Python, PyBullet, GPT-4o

- Built an autonomous navigation agent in which a differential-drive robot perceives through an onboard camera, reasons with a vision-language model (GPT-4o), and executes motor commands in a closed perceive-reason-act loop — a full 0-to-1 robotics system in Python.
- Diagnosed embodiment failures by instrumenting the simulator: proved a world-fixed camera left the robot's heading unobservable, and re-architected to a first-person onboard camera, restoring goal-directed navigation.
- Replaced open-loop actuation with closed-loop control, cutting turn error from ~35% overshoot to within ~2°; caught an inverted-actuator bug and locked control sign conventions with a regression test suite.
- Deliberately split classical vs. learning-based methods — deterministic closed-loop primitives for low-level motion, the VLM for high-level perception and planning — with a real-time visualization + MP4 pipeline to inspect agent behavior.

Autonomous Navigation Controller (TurtleBot4)

Spring 2026

Python, ROS2 Jazzy, Docker, Gazebo

- Engineered an Attractive + Repulsive Potential Field (APF) navigation controller for a simulated TurtleBot4, implementing reactive obstacle avoidance with tunable gain parameters and dynamic goal-seeking behavior.
- Built a ROS2 analysis pipeline using the rosbags library to parse bag files, align optical flow signals across sensor streams, and validate controller performance across multiple test scenarios.
- Containerized the development environment with Docker, enabling reproducible builds and consistent testing across the ROS2 Jazzy stack.

Automated Robotic Control & Perception System

Fall 2025

Python, OpenCV, Inverse Kinematics

- Engineered a Python control API for a 6-DOF commercial robotic arm with closed-loop ArUco marker detection for autonomous object localization, translating Inverse Kinematics into validated physical actuation.

TECHNICAL SKILLS

Robotics, Autonomy & Perception: ROS2, Gazebo, PyBullet, ESP32, Path Planning (A*, APF), Inverse Kinematics, Sensor Fusion, BLE/TCP Networking, OpenCV, YOLO, SIFT/FLANN, ArUco, RANSAC

AI/ML: PyTorch, TensorFlow, Scikit-Learn, Vision-Language Models (VLM), Graph Neural Networks, LLM Agents, RAG

Languages: C++, Python, Java, C, JavaScript/TypeScript, SQL, OCaml

Cloud & DevOps / Tools: AWS (S3), GCP (Vertex AI, BigQuery), Docker, CI/CD, GitHub Actions, Linux, React, FastAPI, Flask, Pandas, NumPy, SciPy